

Probability

Why we Study this?

- One of the most important disciplines in Computer Science (CS).
- Algorithm Design and Game Theory
- Information Theory
- Signal Processing
- Cryptography
- Goal:

To try our best to teach you how to easily and confidently solve problem involving probability

Introduction to probability

Probability is the measure of the likelihood that an event will occur.

- Probability is quantified as a number between 0 and 1 (where 0 *indicates impossibility and 1 indicates certainty*).
- The higher the probability of an event, the more certain we are that the event will occur,
- Example

For an event A the probability of A(also called $P(A)$) will be $P(A) = \frac{\text{The Number of ways event can occur}}{\text{The total Number of possible outcomes}}$

Contents

- Basic definitions and an elementary 4-step process
- Conditional probability and the concept of independence
- Random variables and distributions
- Expected value and deviation from it

Elementary 4-step process

- Finding the sample space
- Defining the events of interest
- Determining outcome probability
- Compute Event Probability

Example:

- Participant is given a choice of three doors.
- Behind one door is a car, behind the others, useless stuff.
- The participant picks a door (say door 1).
- The host, who knows what is behind the doors, opens another door (say door 3) which has the useless stuff.
- He then asks the participant if he would like to switch (pick door 2)?

Step 1: Finding the sample space

- Sample Space:

The sample space S of an experiment is the set of possible outcomes. An event E is a subset of the sample space.

- To find this, we must understand the quantities involved in the random process

Sample Space

Every possible outcome of these quantities is called an outcome.

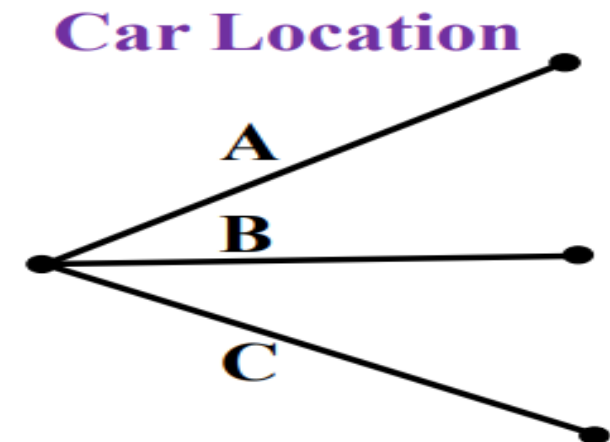
- And (as said earlier) the set of all possible outcomes is called the sample space

A **tree structure (Possibility tree) is a useful tool** for keeping track of all outcomes When the number of possible **outcomes is not too large**

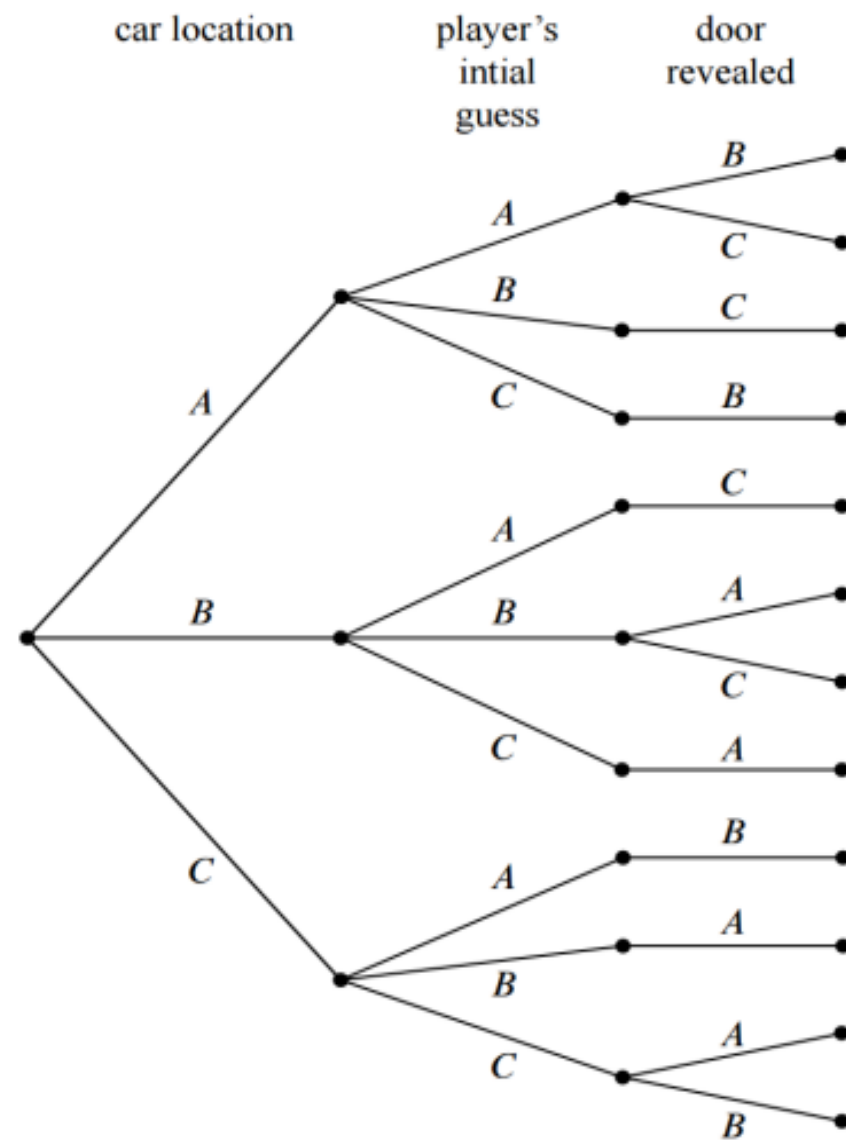
Possibility Tree

The first quantity in our example is the door concealing the car

- Represent this as a root of tree with three branches (three doors)
- For each possible location of the car, the player can choose any of the three doors
- Then the final possibility is regarding the host opening a door to reveal the useless thing



Possible outcomes Tree



Computer Sample Space

- The leaves of the possibility tree represent the outcomes of a random process
- The set of all leaves represent the sample space
- In our example, if we represent the leaves as a sequence of “labels” of intermediate nodes including the leaf node then,
- Sample Space

$$S = \left\{ \begin{array}{l} (A, A, B) (A, A, C) (A, B, C) (A, C, B) (B, A, C) (B, B, A) \\ (B, B, C) (B, C, A) (C, A, B) (C, B, A) (C, C, A) (C, C, B) \end{array} \right\}$$

Step 2: Defining the event of Interest

- Lets say our event of interest is

What is the probability that the car is behind door C?”

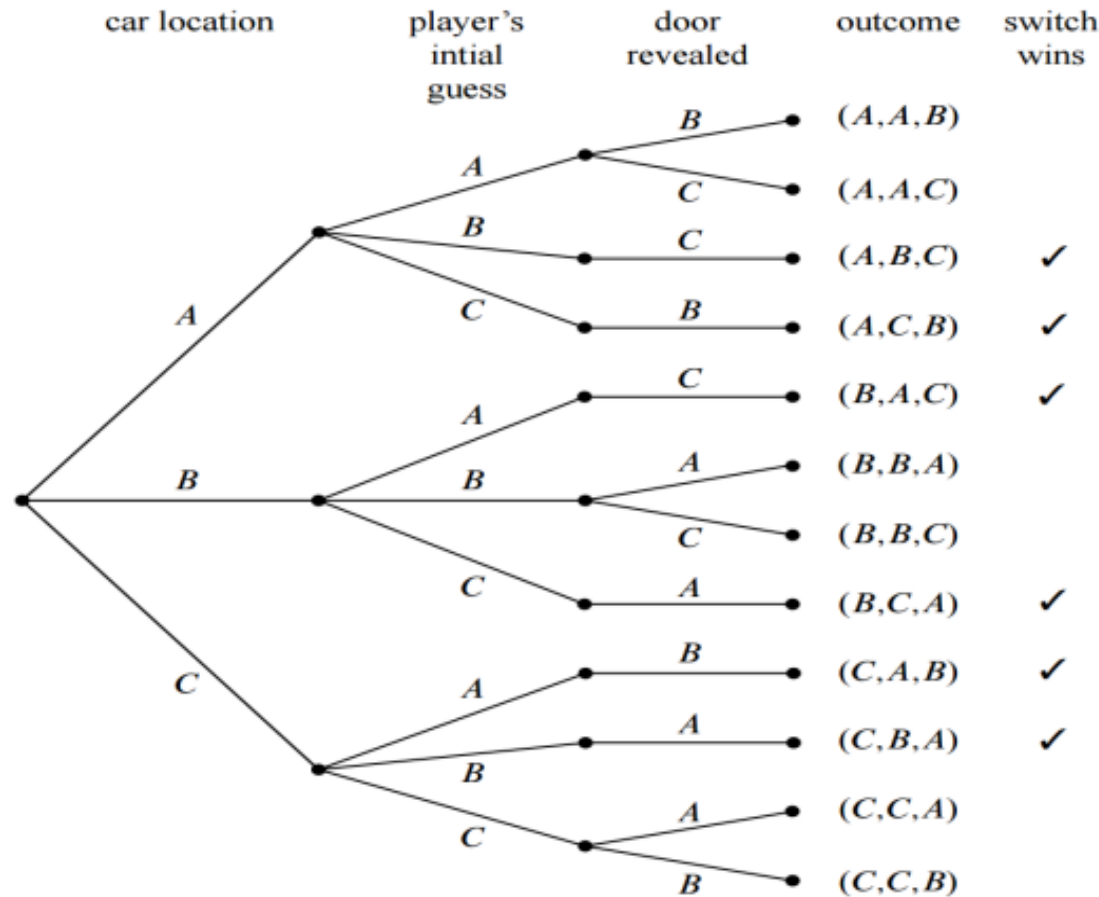
Doing this reduces S to some specific outcomes, called event of interest

The set of possible outcomes reduces to

$(C, A, B)(C, B, A)(C, C, A)(C, C, B)$

Second Event of Interest

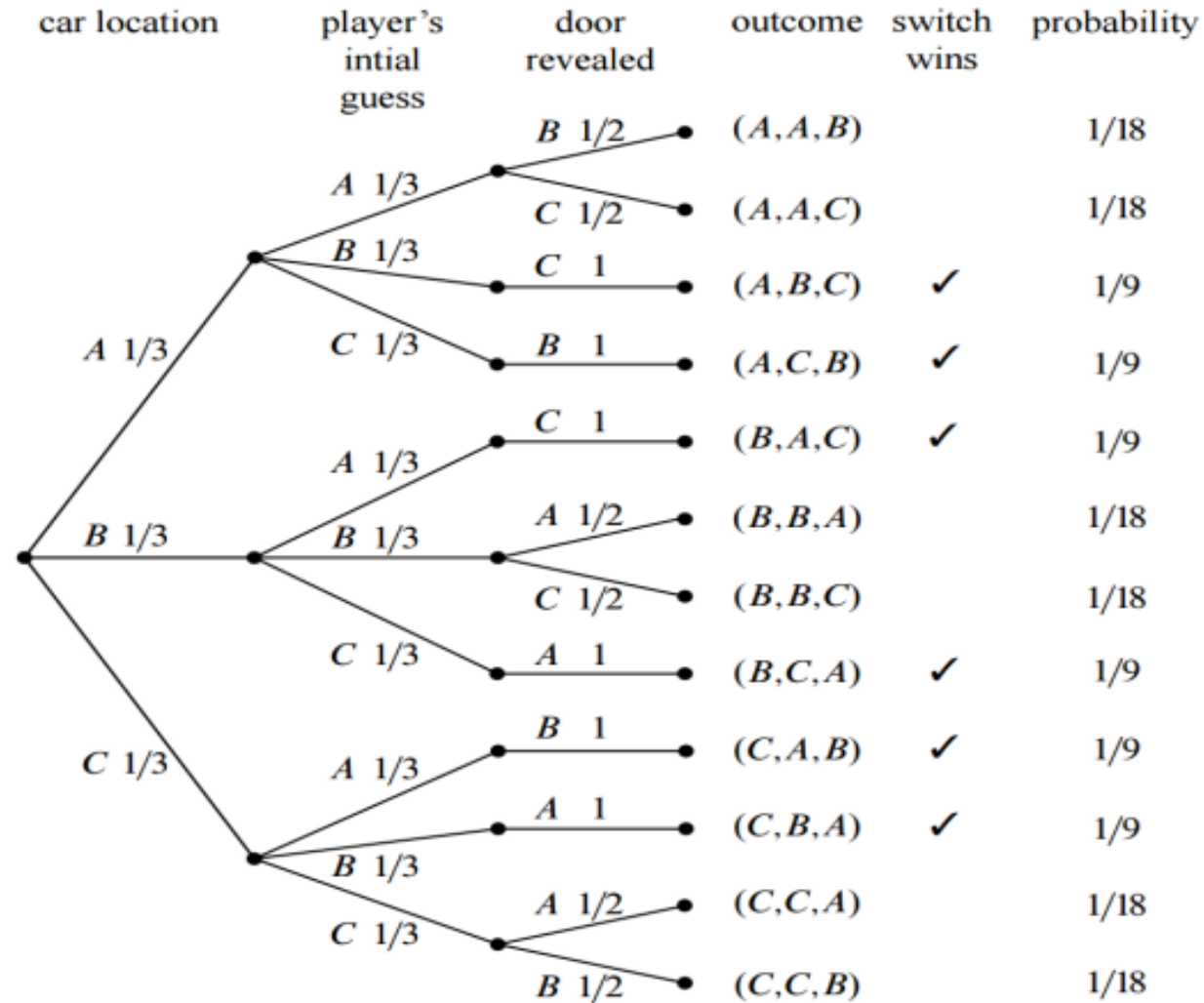
- “What is the probability that the player will win by switching?”



Step 3: Determine outcome Probability

- Assign Edge Probabilities
- Compute Outcome Probabilities

Compute Outcome Probability



Step 4: Compute Event Probability

Pr Switching Wins =

$$Pr(A, B, C) + Pr(A, C, B) + Pr(B, A, C) + Pr[(C, A, B)] + Pr[(C, B, A)]$$

$$= \frac{1}{9} + \frac{1}{9} + \frac{1}{9} + \frac{1}{9}$$
$$= \frac{2}{3}$$

Probability rules

Rules of probability

- Probability addition rule
- Difference Rule
- Inclusion Rule
- Exclusion Rule
- Complement Rule

Addition Rule

- This rule states that the number of elements in a union of mutually disjoint finite sets equals the sum of the number of elements in each of the component sets.

$$N(A) = N(A_1) + N(A_2) + \dots + N(A_k)$$

- Example

A computer access password consists of from one to three letters chosen from the 26 in the alphabet with repetitions allowed. How many different passwords are possible?

Solution:

- By the addition rule, the total number of passwords equals the number of passwords of length 1, plus the number of passwords of length 2, plus the number of passwords of length 3.

Number of passwords of length 1 = 26

Number of passwords of length 2 = 26^2

Number of passwords of length 3 = 26^3

$$\begin{aligned}\text{Total no of passwords} &= 26 + 26^2 + 26^3 \\ &= 18278\end{aligned}$$

Difference Rule

- if the number of elements in a set A and the number in a subset B of A are both known,

Number of elements that are in A and not in B can be computed.

The Difference Rule:

If A is finite set and B is a subset of A, then

$$N(A-B) = N(A) - N(B)$$

Example

- Example:

A typical PIN (personal identification number) is a sequence of any four symbols chosen from the 26 letters in the alphabet and the ten digits, with repetition allowed.

1. How many PINs contain repeated symbols?
2. If all PINs are equally likely, what is the probability that a randomly chosen PIN contains a repeated symbol?

- Part A
- The total number of possible PINs is the number of choices for each symbol raised to the power of the number of symbols in a PIN.
- In this case, there are 36 choices (26 letters + 10 digits) for each symbol, and a PIN consists of 4 symbols.

$$36^4 = 1,679,616.$$

